SCIENCE AND TECHNOLOGY COUNCIL
IIT(BHU) VARANASI

INTER IIT TECH MEET 15.0

PREPATHON PS

ELECTRONICS (FRONTEND VLSI)

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8-Channel Polyphase Channelizer

Inter IIT Tech Meet 15.0 Prepathon — Frontend VLSI

INTRODUCTION

Wideband receivers frequently digitize several independent channels in a single stream. Extracting each channel with a dedicated direct-form FIR filter requires a large number of arithmetic operations per input frame of M samples. A polyphase filter bank followed by an FFT shares the filtering across all channels, reducing arithmetic cost by exploiting the decimation factor and FFT symmetries.

The problem is to implement a fixed-size 8-channel polyphase channelizer in synthesizable RTL and to evaluate the relationship between arithmetic hardware and per-sample cycle count. A fully parallel implementation sustains one sample per cycle at larger area. A folded implementation time-shares a smaller number of multiply-accumulate units, sustaining a lower rate at reduced area.

PROBLEM STATEMENT DESCRIPTION

Design, verify and synthesize an 8-channel, critically sampled polyphase channelizer (polyphase FIR followed by an 8-point FFT, or any mathematically equivalent structure) that satisfies the numerical specification below, meets timing at 100 MHz in a Nangate 45nm flow. Definition of the streaming, configuration and handshake interfaces is left to the participant.

Specification

Parameter	Value
Channels M	8 (critically sampled, decimation by 8)
Input	Real, 16-bit signed integer samples
Prototype filter	$L = 32$ taps (4 taps per branch), 16-bit signed Q1.15 coefficients, arbitrary values (symmetry may not be assumed)
Output	8 complex bins per frame, 16-bit signed values, delivered serially in increasing channel order
Frame rate	One frame per 8 input samples

Mathematical Definition

For frame m (newest input sample $x[8m+7]$) and channel $k \in [0, 7]$:

$$Y_{k[m]} = \sum_{n=0}^{31} h[n] e^{-j2\pi kn/8} x[8m + 7 - n]$$

Equivalently, with branch outputs $v_{r[m]} = \sum_{j=0}^{3} h[8j + r] x[8(m - j) + 7 - r]$ for $r = 0, ..., 7$:

$$Y_{k[m]} = \sum_{r=0}^{7} v_{r[m]} e^{-j2\pi kr/8}$$

Numerical Specification

Each output bin produced by your Verilog must agree with the ideal Python reference model (computed via a double-precision floating-point FFT) to within **±2 LSB**.

What does ±2 LSB mean?

Because hardware requires fixed-point arithmetic (e.g., truncating or rounding fractional bits after a multiplier to save area), your final hardware outputs will inherently suffer from quantization noise. The ±2 LSB limit means your Verilog integer output is allowed to deviate by a maximum numerical value of 2 from

the mathematically perfect Python integer output. If your error is greater than 2, you have aggressively truncated too many bits internally and failed the precision requirement.

- **FIR stage:** exact integer accumulation, scaled by 2^{-15}, rounded to nearest even and saturated to 16 bits to produce v_r.
- **FFT stage:** the ideal Python reference outputs are computed with a double-precision floating-point FFT. Y_k is scaled by $\frac{1}{8}$, rounded to nearest even and saturated to 16 bits.
- **Implementation freedom:** the FIR stage must be exact before scaling. FFT word lengths and twiddle precision are design parameters, provided the ±2 LSB bound is met.
- **Initialization:** the first 3 frames after reset are excluded from value checking, but must still produce the correct output transfers.

Scoring

- Timing constraints: **10.0 ns clock period (100 MHz)**. Setup and hold must be met. Multicycle paths are not permitted.
- Timing will be validated using OpenSTA and the Nangate 45nm library.

METHODOLOGY

The following methodology is recommended.

1. **Complexity analysis:** derive the arithmetic cost (real multipliers and adders per frame) of 8 independent direct-form channels and of the polyphase FIR with FFT for $M = 8$, $L = 32$. State the counting convention, including real-by-complex products and trivial twiddle factors.
2. **Reference model:** implement the numerical specification in Python (integer FIR with round-half-to-even and saturation, double-precision FFT, output rounding and saturation) and export stimulus and expected-output `.hex` files.
3. **Fixed-point FFT analysis:** determine the minimum datapath and twiddle word lengths that guarantee the ±2 LSB bound, supported by an analytical worst-case error bound and randomized fixed-point simulation.
4. **Architecture exploration:** select the multiplier count, folding factor and pipeline depth. Justify your pipeline and folding architecture choices to ensure timing closure at 100 MHz.
5. **RTL implementation:** implement the polyphase FIR, FFT stage, and serial output stage in synthesizable Verilog or SystemVerilog.
6. **Verification:** develop a self-checking testbench covering single-tone, multi-tone, and noise stimuli, reporting `PASS` or `FAIL`.
7. **Synthesis and timing closure:** develop local Yosys and OpenSTA scripts and constraints for Nangate 45nm, close timing at 100 MHz with margin, and report timing slack and latency.

Software

- Icarus Verilog or Verilator
- Yosys and OpenSTA (open-source tools only)
- Nangate 45nm standard-cell library
- Python 3

PHASE 1 — MID-TERM

Deadline: 8 October 2026, EOD

Tasks

1. **Complexity derivation:** comparison of direct-form and polyphase-plus-FFT structures as described in Methodology step 1.
2. **Fixed-point FFT study:** Python reference model and minimum bit-widths with a worst-case error justification.
3. **Architecture proposal:** RTL datapath diagram, target multiplier count, with justification of the pipeline depth required to meet timing.

Deliverables

- **M1:** Report (PDF, maximum 4 pages)
- **M2:** Python reference model and stimulus generator

PHASE 2 — END-TERM

Deadline: 12 October 2026, EOD

Deliverables

1. `rtl/` — synthesizable sources
2. `tb/` — Python reference model, stimulus generator and self-checking testbench
3. `synth/` — synthesis scripts (`.tcl`, `.sdc`) and timing reports
4. `report.pdf` — Phase 1 content and results dashboard

Submitted as a single ZIP archive.

Results Dashboard

1. Static Timing Analysis (STA) report showing positive slack at 100 MHz
2. Latency, in cycles, from the newest input sample of a frame to the first output bin of the same frame
3. Worst-case measured error (LSB) over the test suite

Verification Checkpoints

Hint on Verification Flow: *Teams must write a Python reference model to mathematically generate the digital input signals and calculate their ideal expected outputs, exporting both to `.hex` files. The Verilog testbench will then feed the inputs into the RTL simulation and automatically compare the hardware's output against the expected outputs to report PASS or FAIL.*

ID	Checkpoint	Signal Generation & Pass Criterion
V1	Single-tone	**Generation:** In Python, generate a pure digital sine wave whose frequency exactly matches the center of a specific channel k. **Criterion:** The signal must appear predominantly in bins k and $8-k$. All output bins must match the ideal Python FFT reference within ±2 LSB.
V2	Multi-tone	**Generation:** Add multiple pure sine waves of varying frequencies together. **Criterion:** The hardware's response must match the ideal Python FFT reference on all output bins within ±2 LSB.
V3	Multi-tone with noise	**Generation:** Superimpose random uniform noise (e.g., using `numpy.random`) onto the multi-tone signal. **Criterion:** The hardware's response must match the ideal Python FFT reference on all output bins within ±2 LSB.

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EVALUATION CRITERIA

1. **Functional correctness:** V1, V2, and V3 pass within ±2 LSB.
2. **Timing Closure:** Successfully meeting the 10.0 ns (100 MHz) constraint in Yosys/OpenSTA.
3. **Verification quality:** fidelity of the reference model and coverage of the testbench.
4. **Reporting:** accuracy of the dashboard, latency, and architectural justification.

JUDGING CRITERIA

Mid-Term (30 Marks)

Criterion	Marks
Complexity derivation (direct form vs polyphase + FFT)	10
Fixed-point FFT study and reference model correctness	15
Architecture clarity and justification	5
Total	**30**

End-Term (70 Marks)

Criterion	Marks
Functional correctness (V1, V2, & V3)	45
Timing Closure (Positive Slack at 100 MHz)	10
Verification quality and testbench rigor	10
Results dashboard and latency	5
Total	**70**

SUBMISSION

Submission
Submit the completed problem statement deliverables via the submission form: https://forms.gle/9hxci4r4fZMKywzR8. **Phase 1 — Mid-Term:** 8 October 2026, EOD. **Phase 2 — End-Term:** 12 October 2026, EOD.